ePolicies AI Assistant – Run & Update Guide (Windows laptop)

All commands for starting, stopping, updating and fixing the app. Type each command in **PowerShell** (the VS Code terminal is fine), one line at a time.

1. Where everything is

What	Location
Project folder	`C:\Users\n692121\Downloads\epolicies-rag-main\epolicies-rag-main`
Policy `.md` files	`...\epolicies-rag-main\output`
Scraper (downloads policies from the website)	`...\epolicies-rag-main\scrape_epolicies.py`
Python environment	`C:\Users\n692121\Downloads\epolicies-rag-main\.venv` (one folder above the project)
Backend (Python API)	`...\epolicies-rag-main\backend`
Website (Next.js)	`...\epolicies-rag-main\frontend`
Settings (keys, database, model)	`...\epolicies-rag-main\.env`
Saved embeddings (backup of the index)	`...\epolicies-rag-main\artifacts\embedding.jsonl`
Database (your own PostgreSQL 16)	`C:\Users\n692121\epolicies-pgdata`, port **5433**
Website address	**http://localhost:3000**
Backend address	http://localhost:8000

2. Every day – start and stop (one click)

Start: double-click **`start-windows.bat`** in the project folder. It:

1. starts the database (if it is not running),
2. checks the Google Cloud login and opens a sign-in page if it expired,
3. runs `doctor` (health check),
4. opens two windows: **"ePolicies backend"** and **"ePolicies website"**,
5. opens http://localhost:3000 in the browser.

Keep the two windows open while you use the app.

Stop: double-click **`stop-windows.bat`**. It stops the website, the backend and the database.

> If Windows says "Windows protected your PC": click **More info** → **Run anyway**.

3. Manual start (if the .bat file does not work)

Open **three** terminals.

Terminal 1 – database:

```
& "C:\Program Files\PostgreSQL\16\bin\pg_ctl.exe" -D "$HOME\epolicies-pgdata" -o "-p 5433"
-l "$HOME\epolicies-pgdata\log.txt" start
```

Terminal 1 – Google Cloud login check (only if Gemini fails):

```
gcloud auth application-default login
gcloud auth application-default set-quota-project hcb-dev-genai-tools
```

Terminal 2 – backend:

```
cd C:\Users\n692121\Downloads\epolicies-rag-main\epolicies-rag-main\backend
..\..\.venv\Scripts\Activate.ps1
python -m app.cli doctor
uvicorn app.main:app --port 8000
```

Terminal 3 – website:

```
cd C:\Users\n692121\Downloads\epolicies-rag-main\epolicies-rag-main\frontend
npm run dev
```

Open **http://localhost:3000**.

Manual stop: press **Ctrl+C** in terminals 2 and 3, then stop the database:

```
& "C:\Program Files\PostgreSQL\16\bin\pg_ctl.exe" -D "$HOME\epolicies-pgdata" stop
```

4. Every week – update the policies from the website

The website's policies change every week. Updating has two parts: **(A) the scraper downloads the latest pages into `output`**, then **(B) the app re-indexes only what changed.**

One click: double-click **`update-windows.bat`** in the project folder (next to `scrape_epolicies.py`). It does steps 1–4 below for you: starts the database if needed, checks the Google login, backs up `output` and `embedding.jsonl`, runs the scraper, shows what changed, **asks Y/N**, then updates the app. The manual steps are below if you need them.

Before you start: the database must be running (start-windows.bat or section 3). Close Microsoft Edge (the scraper uses your Edge profile).

Step 1 – go to the project folder

```
cd C:\Users\n692121\Downloads\epolicies-rag-main\epolicies-rag-main
```

Step 2 – back up the current files

```
Copy-Item output output_backup -Recurse -Force
Copy-Item artifacts\embedding.jsonl artifacts\embedding-backup.jsonl
```

Step 3 – download the latest policies (A)

```
python scrape_epolicies.py --update
```

First time only, test on 2 topics: `python scrape_epolicies.py --update --limit 2`

Step 4 – update the app (B)

```
cd backend
..\..\.venv\Scripts\Activate.ps1
python -m app.cli check-files
python -m app.cli ingest --dry-run
python -m app.cli ingest
python -m app.cli status
```

- `check-files` – offline check of every `.md` file (free).
- `ingest --dry-run` – shows **new / modified / unchanged / deleted** counts; changes nothing. **If it says many files are deleted, stop** – the scraper probably failed.
- `ingest` – sends only new and changed files to Gemini; updates `embedding.jsonl` at the end.
- `status` – last run should say **completed** with **0 failed**.

No restart needed: the website uses the new data on the next question.

Scraper options (from `python scrape_epolicies.py --help`)

Option	Meaning
`--update`	Update existing files (use this every week)
`--limit N`	Only process the first N topics (for testing)
`--output FOLDER`	Output folder (default: `output`)
`--user-data-dir` / `--profile`	Which Edge profile to use (defaults are fine)

Scraper needs these libraries (one-time install):

```
python -m pip install beautifulsoup4 markdownify playwright
python -m playwright install chromium
```

5. Recommended settings in `.env`

Open `.env` in the project folder, set these lines (add any that are missing), save, then restart the backend (stop-windows.bat, then start-windows.bat).

```
GEMINI_MODEL=gemini-2.5-flash
GEMINI_TEMPERATURE=0.1
GEMINI_THINKING_BUDGET=1024
TOP_K_FINAL=12
MAX_CONTEXT_CHUNKS=16
MAX_CONTEXT_CHARS=40000
SYNONYM_EXPANSION=true
PLANNER_MODE=auto
```

Setting	Why
`GEMINI_TEMPERATURE=0.1`	Small variation in wording between answers; the facts still come only from the cited policies
`GEMINI_THINKING_BUDGET=1024`	Gemini reasons before answering (better answers); if "malformed JSON" errors come back, use 512
`TOP_K_FINAL` / `MAX_CONTEXT_*`	More passages per answer, so broad questions are more complete
`SYNONYM_EXPANSION=true`	"abortion" also finds "termination of pregnancy", acronyms like COB match their full names

6. Useful app commands

Run inside `backend` with the environment active (`..\..\.venv\Scripts\Activate.ps1`).

Command	What it does
`python -m app.cli doctor`	Health check: files, database, Gemini
`python -m app.cli status`	How many documents/chunks are indexed, last runs
`python -m app.cli check-files`	Offline check of all `.md` files
`python -m app.cli ingest --dry-run`	Show what would change (writes nothing)
`python -m app.cli ingest`	Index new/changed files
`python -m app.cli resume`	Continue an ingest that stopped halfway
`python -m app.cli ingest --retry-failed`	Retry only files that failed
`python -m app.cli ask "your question"`	Ask a question in the terminal
`python -m app.cli search "words" --mode keyword`	See which chunks a search finds
`python -m app.cli export-jsonl`	Re-write `embedding.jsonl` from the database
`python -m app.cli load-jsonl`	Rebuild the database from `embedding.jsonl` (no Gemini cost)
`python -m app.cli reconcile`	Compare `embedding.jsonl` with the database

7. Troubleshooting

What you see	Fix
Website: *"Could not open a PostgreSQL connection pool"*	Database is stopped. Run start-windows.bat or the `pg_ctl ... start` command (section 3).
doctor: `[FAIL] gemini-embedding-001 ... credentials were rejected or expired`	`gcloud auth application-default login`, then restart the backend.
doctor: `[WARN] pgvector is not installed`	Normal on this laptop – the app uses its built-in fallback. Nothing to do.
`pg_ctl: could not start server`	Read the log: `Get-Content "$HOME\epolicies-pgdata\log.txt" -Tail 20`
Port 8000 or 3000 already in use	Run stop-windows.bat, or find the process: `netstat -ano | findstr 8000` then `taskkill /PID <number> /F`
`ModuleNotFoundError: No module named '...'`	`python -m pip install <name>` (type it yourself, no long path in front)
`No such file or directory`	You are in the wrong folder – `cd` to the project folder first (section 1).
`.ps1 cannot be loaded because running scripts is disabled`	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` (once)
Website red "hydration" message	Get the latest `theme-toggle.tsx` from GitHub; refresh with Ctrl+Shift+R.
An answer takes ~30–40 seconds	Normal – most of the time is Gemini (planning, answer, verification).

8. Undo a bad weekly update

```
cd C:\Users\n692121\Downloads\epolicies-rag-main\epolicies-rag-main
Remove-Item output -Recurse -Force
Copy-Item output_backup output -Recurse
cd backend
..\..\.venv\Scripts\Activate.ps1
python -m app.cli load-jsonl --path ..\artifacts\embedding-backup.jsonl --allow-unverified
--prune
```

This puts back last week's files and index without any Gemini cost.

9. When IT installs pgvector (later)

Use a **new** database so the current one stays as a backup:

```
& "C:\Program Files\PostgreSQL\16\bin\createdb.exe" -p 5433 -U postgres epolicies_pgv
& "C:\Program Files\PostgreSQL\16\bin\psql.exe" -p 5433 -U postgres -d epolicies_pgv -c
"CREATE EXTENSION vector;"
```

In `.env`, change the end of `DATABASE_URL` from `/epolicies` to `/epolicies_pgv`, then in `backend`:

```
python -m app.cli init-db
python -m app.cli load-jsonl
python -m app.cli doctor
```

The pgvector warning disappears. No Gemini cost. To go back, change `DATABASE_URL` back.

> Do **not** use `python -m app.cli rebuild --confirm` for this – it deletes all embeddings and re-creates them with Gemini (API cost).

10. Getting new code from GitHub

Repository: **https://github.com/charan4590/epolicies-rag**

- **One file:** open it on GitHub → **Download raw file** → put it in the same folder on the laptop.
- **Everything:** green **Code** → **Download ZIP** → extract. Then copy these from the old folder into the new one (they are not on GitHub): `.env`, `output\`**, **`artifacts\embedding.jsonl`**, **`scrape_epolicies.py`**, **`show_diff.py`**, **`verify_output.py`**. Run **`setup-windows.bat`** once in the new folder.